INCH-POUND

MIL-A-82832(OS)
5 February 1990
SUPERSEDING
WS 20643A
9 December 1986

MILITARY SPECIFICATION

ADHESIVE, EPOXY RESIN, STRUCTURAL BONDING

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of adhesive consisting of a filled epoxy resin and an amine curing agent, referred to herein as the adhesive.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1340

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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STANDARDS

FEDERAL

FED-STD-313 Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 **Other Government documents, drawings and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFRs)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 214 Sieve Analysis of Granular Metal Powders (DoD adopted)

ASTM D 740 Methyl Ethyl Ketone (DoD adopted)

ASTM D 1002 Strength Properties of Adhesives in Shear by Tension Loading (Metal-to-Metal) (DoD adopted)

ASTM D 1652 Epoxy Content of Epoxy Resins (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample of the adhesive shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 **Material.** The adhesive shall be a two-part adhesive consisting of a gray, aluminum powder filled, thixotropic epoxy resin in paste form (part A) and an amber, amine-type curing agent in liquid form (part B) in accordance with the following:

a. Part A 100 ± 1 parts by weight

b. Part B 33 ± 1 parts by weight

3.2.1 **Aluminum powder particle size.** The particle size of the aluminum powder used as filler material shall be such that a minimum of 93 percent of the powder will pass through a number 325 mesh sieve.

3.2.2 **Kits.** The adhesive shall be supplied in kits consisting of separate containers for part A and part B.

3.3 **Chemical and physical properties.** Chemical and physical properties of the adhesive shall conform to the requirements specified in table I.

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TABLE I. Chemical and physical properties.

Property	Unit	Ingredients				Mixed Material	
		Part A		Part B			
		Min.	Max.	Min.	Max.	Min.	Max.
Filler content	percent	40	47	---	---	---	---
Epoxide content	g/equiv.	160	200	---	---	---	---
Titratable nitrogen	percent	---	---	11.8	14.0	---	---
Tensile adhesion strength	psi	---	---	---	---	2400	---
Working life	minutes	---	---	---	---	40	---

3.4 **Safety.**

3.4.1 **Toxic products and formulations.** The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.4.2 **Material Safety Data Sheet (MSDS).** The contractor shall prepare and submit a MSDS in accordance with FED-STD-313 as specified in the contract (see 6.2).

3.5 **Shelf life.** The adhesive shall have a shelf life of 12 months after the date of manufacture when stored at 40 degrees Fahrenheit (°F) maximum in the original, closed containers.

3.5.1 **Shelf life extension.** The shelf life expiration date of an individual lot may be extended for additional 12 month periods provided the adhesive, upon retest, conforms to the tensile adhesion and working life requirements specified in table I.

3.6 **Workmanship.** The adhesive component materials shall be free from agglomeration, foreign matter, separation, dilution, and contamination.

4. **QUALITY ASSURANCE PROVISIONS**

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use

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his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. the absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Toxicological product formulations. The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the quality conformance inspection specified in 4.4. Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of examination and tests specified in table II.

4.4.1 Sampling for inspection.

4.4.1.1 Lot. For inspection purposes, a lot of adhesive shall consist of the material of each part compounded and mixed in a single container in one mixing period without change in process or materials and subject to inspection at one time.

4.4.1.2 Sample material. Sufficient material shall be selected at random from each lot to perform all of the inspections specified in 4.5.1 through 4.5.4.

4.4.1.3 Noncompliance of samples. Failure of one or more of the sample containers of material to meet all of the specified requirements shall be cause for rejection of the lot.

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TABLE II. Quality conformance inspection.

Examination and tests	Requirement	Test method
Visual examination	3.6 and section 5	4.5.1
Filler content (Part A)	3.3	4.5.2
Epoxide content (Part A)	3.3	4.5.3
Titratable nitrogen (Part B)	3.3	4.5.4
Tensile adhesion strength (mixed material)	3.3	4.5.5
Working life (mixed material)	3.3	4.5.6
Aluminum powder particle size	3.2.1	4.5.7

4.5 Methods of inspection. The following inspection methods shall be used. Unless otherwise specified in the method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all inspections defined herein. Solvents and indicators may be commercial nonreagent materials.

4.5.1 Visual examination. Each sample of material shall be examined visually for conformance to section 3.6 and section 5.

4.5.2 Filler content (part A). Filler content of part A shall be determined in accordance with the following:

a. Weigh 0.7 to 0.8 gram (g) of sample, to the nearest 0.1 milligram (mg), into a beaker, and add 10 to 20 milliliters (ml) of chlorobenzene. (To prevent solids from plugging up the filter, 0.35 to 0.40 g of filter aid (see 6.5) that has been previously dried for 60 ± 12 minutes at 283 to 302°F may be added.) Stir until all lumps are dissolved and any suspended insoluble matter is finely dispersed.

b. Filter the sample through a tared, medium-porosity, sintered-glass crucible into a vacuum flask, using vacuum. Wash the residue with 50 ± 10 ml of chlorobenzene. Retain the filtrate for the epoxide content determination (see 4.5.3).

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c. Dry the crucible and contents for a minimum of 30 minutes at 300 ± 10°F. Place in a desiccator, cool to ambient temperature, and weigh to the nearest 0.1 mg.

d. Calculate the filler content as follows:

(1) Filler content (percent) =

$$\frac{(W1 - F)}{W} \times 100 \text{ (without filter aid)}$$

(2) Filler content (percent) =

$$\frac{W1 - (F + C)}{W} \times 100 \text{ (with filter aid)}$$

Where: W1 = weight of crucible and residue, g

W = weight of sample,

F = weight of crucible,

C = weight of filter aid,

e. Record the filler content to the nearest 0.01 percent.

f. Record the average of a minimum of two determinations.

4.5.3 **Epoxide content (part A).** Epoxide content of part A shall be determined in accordance with ASTM D 1652. (If the filler content has been determined, the filtrate may be used for the sample).

4.5.4 **Titratable nitrogen (part B).** Titratable nitrogen of part B material shall be determined in accordance with the following:

4.5.4.1 **Equipment.**

a. Calomel reference electrode.

b. Glass electrode, all purpose.

c. pH meter.

4.5.4.2 **Reagents.**

a. Acetic acid, glacial, American Chemical Society (ACS) reagent grade.

b. Acetic anhydride, ACS reagent grade.

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c. Potassium acid phthalate, primary standard.

d. Perchloric acid, ACS reagent grade.

4.5.4.3 **Perchloric acid solution.** Perchloric acid solution, 0.1 Normal (N), shall be prepared as follows:

a. Place 250 ± 50 ml of glacial acetic acid in 1000-ml volumetric flask.

b. Add 8 to 9 ml of 70-percent perchloric acid or 10 to 11 ml of 60-percent perchloric acid, and mix.

c. Add 20 ml of acetic anhydride, and mix.

d. Dilute to volume with glacial acetic acid, and mix.

e. Stopper flask, and let stand for minimum of 24 hours.

f. Weigh 0.45 to 0.50 g of dried potassium acid phthalate, to nearest 0.1 mg, into 250-ml beaker.

g. Add 50 to 100 ml of glacial acetic acid. Stir and heat gently until all of potassium acid phthalate has dissolved.

h. Cool to room temperature, immerse the electrodes in sample (solution), and titrate potentiometrically with perchloric acid.

i. Perform blank determination.

j. Calculate as follows:

$$\text{Normality of perchloric acid} = \frac{(W)\ (4.897)}{V1 - V2}$$

Where: W = weight of potassium acid phthalate,

V1 = volume of perchloric acid required to titrate sample, ml

V2 = volume of perchloric acid required to titrate blank, ml

4.897 = reciprocal of milliequivalent weight of potassium acid phthalate

k. Normality between duplicate standardizations shall not differ more than 0.001.

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4.5.4.4 Procedure. Titratable nitrogen content shall be determined as follows:

a. Weigh to the nearest 0.1 mg a 2 to 4 milliequivalent sample into a 250-ml beaker.

b. Add 50 to 100 ml of glacial acetic acid. If necessary, warm the solution to completely dissolve the sample.

c. If warming was used to dissolve the sample, allow the solution to cool to room temperature.

d. Place the electrodes in solution, and titrate potentiometrically with standardized perchloric acid solution.

e. Perform a blank determination.

f. Calculate as follows:

$$\text{Titratable nitrogen (percent)} = \frac{(V1 - V2)\ (N)\ (0.014)}{W} \times 100$$

Where: V1 = volume of perchloric acid solution required to titrate sample, ml

V2 = volume of perchloric acid solution required to titrate blank, ml

N = normality of perchloric acid solution

W = weight of sample,

0.014 = milliequivalent weight of nitrogen.

g. Record the average of a minimum of two determinations.

4.5.4.5 Alternative titration procedure. Alternatively, the standardization and the sample titration may be performed visually using crystal violet indicator, providing the sample coloration does not interfere with the observation of the blue-green endpoint. In case of dispute, the potentiometric titration procedure shall govern.

4.5.5 Bond shear strength. The bond shear strength of the adhesive shall be determined in accordance with ASTM D 1002 and the following:

a. Prepare four test panels, 8 inches long by 4 inches wide by 0.063 $\pm$ 0.005 inch thick, from clad 2024-T3 aluminum alloy.

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b. Degrease the panels with methylene chloride, methyl chloroform, or methyl ethyl ketone conforming to ASTM D 740. Allow to dry in air for a minimum of 45 minutes.

c. Prepare a solution consisting of 1 liter of deionized water, 300 g of sulfuric acid, 60 grams of sodium dichromate, and 1.5 g of 2024 bare aluminum. Maintain the solution at 150 to 160°F.

d. Place the panels into the solution, and allow to remain for 12 to 15 minutes.

e. Remove the panels from the solution, immediately spray rinse with tap water for 5 minutes, and soak rinse in a deep bath of tap water for a minimum of 5 minutes.

f. Remove the panels, and examine for water breaks. If water breaks are observed, repeat steps b through e.

g. Dry the panels for 30 minutes minimum in an oven maintained at 120 to 140°F.

h. Remove the panels from the oven, and allow to cool to ambient temperature.

i. Mix the adhesive in a ratio of 100 ± 1.0 parts by weight of part A to 33 ± 1.0 parts by weight of part B.

j. Apply a layer of adhesive 0.005 ± 0.001 inch thick to the panels in the area to be bonded.

k. Assemble the test panels in pairs, with an adhesive overlap of 0.500 ± 0.005 inch, and secure rigidly. Apply sufficient pressure to ensure an adhesive thickness of 0.005 ± 0.001 inch.

l. Cure the assembled panels for a minimum of 2 hours at 200 ± 5°F and cool to room temperature.

m. Cut five test specimens from each panel assembly. (Discard a minimum of 0.25 inch of the joint from each end of the assembly.)

n. Determine the shear area by measuring, to the nearest 0.005 inch, the width of the specimen and the length of the overlap.

o. Record the adhesive bond strength of each specimen to the nearest psi. Record the percent adhesive or cohesive failure.

4.5.6 **Working life (mixed material).** Working life of the mixed material shall be determined in accordance with the following:

a. Condition Part A and Part B at 77 ± 5°F for minimum of 10 hours.

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b. Thoroughly mix the adhesive in a ratio of 100 ± 1 parts by weight (pbw) of Part A to 33 ± 1 pbw of Part B and in sufficient quantity to provide 400 to 500 g total sample weight. Stir vigorously with spatula until a uniform consistency is obtained.

c. Start timing at the completion of mixing.

d. Allow the sample to stand undisturbed in environment of 77 ± 5°F until tested for working life.

e. Test for spreadability to a uniform thickness of 0.05 ± 0.01 inch with a spatula at 40 minutes after start of timing. If spreadable, record adhesive as "conforming". If not spreadable, record adhesive as "nonconforming".

f. Record results for a minimum of one determination.

4.5.7 **Aluminum powder particle size.** Particle size of the aluminum powder filler shall be determined by sieve analysis in accordance with ASTM B 214 except that only a number 325 mesh sieve shall be used.

4.6 **Inspection of packaging.** The adhesive containers and container markings shall be visually examined to verify compliance with section 5.

5. PACKAGING

5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 **Level C.** The adhesive shall be packaged to afford adequate protection against loss, contamination, deterioration and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.5. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulation as applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, each container marking shall include the following information:

a. Title, number and date of this specification.

b. Manufacturer's name and address.

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c. Material trade name and identification.

d. Net weight of contents.

e. Lot number and date of quality conformance inspection.

f. Shelf life and inspection/test date for extension (see 3.5.1). In addition, the marking shall include the following:

"SHELF LIFE EXTENSION

THE SHELF LIFE OF AN INDIVIDUAL LOT MAY BE EXTENDED FOR AN ADDITIONAL 12 MONTH PERIOD PROVIDED THE ADHESIVE, UPON RETEST, CONFORMS TO THE REQUIREMENTS SPECIFIED IN MIL-C-82832."

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The adhesive specified herein is intended for general use in structural bonding of components in the Standard Missile MK 104 Dual Thrust Rocket Motor.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2 and 2.2).

c. Whether first article is required (see 3.1 and 6.3).

d. Material Safety Data Sheets (see 3.4.2 and 6.4).

e. Special marking if other than as specified in 5.2.2.

6.3 First article. When first article inspection is required, the contracting officer should provide specified guidance to offerors on the material to be inspected. The material should be a first article sample and should consist of a quantity of the adhesive sufficient to perform all of the inspections specified in 4.3. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such

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production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 **Material Safety Data Sheet (MSDS).** MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR), Part 23, Subpart 23.3. Contracting officers will identify those activities requiring copies of completed MSDSs prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.5 **Possible material.** The following material has been found to meet the filter aid requirements specified in 4.5.2. This is only given for information and is not restrictive.

Celite - manufactured by Johns-Manville

6.6 **Subject term (key word) listing.**

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

Preparing Activity:
NAVY-OS
(Project 1340-N151)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-A-82832	2. DOCUMENT DATE (YYMMDD) 5 February 1990

3. DOCUMENT TITLE
ADHESIVE, EPOXY RESIN, STRUCTURAL BONDING

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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